

Scalability & Future Plan

Date	02 July 2026
Team ID	SWTID-2026-3218
Project Name	Credit Card Approval System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single-server deployment with no load balancing	Cannot handle a high volume of concurrent credit card applications	High
2	No caching layer for repeated approval-check queries	Slower response times as user base grows	Medium
3	Model retraining is done manually	Delayed adaptation to new applicant/risk patterns	Medium

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports around 100 concurrent users on a single instance	Deploy on cloud (AWS/Azure) with auto-scaling and load balancers to support 10,000+ concurrent users
2	Data Storage	Local SQL database with limited capacity	Migrate to scalable cloud storage (Amazon RDS / Azure SQL) with partitioning and backups
3	Performance	Synchronous request processing causing delays under load	Introduce asynchronous processing and a Redis caching layer to reduce latency
4	Security	Basic username/password authentication	Implement OAuth 2.0, data encryption in transit/at rest, and role-based access control

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Real-time integration with credit bureau APIs	04/07/2026	Faster and more accurate approval decisions
Phase 3	Mobile app support and multi-language UI	04/07/2026Q3 2026	Wider user reach and improved accessibility
Phase 4	AI-based fraud detection module	04/07/2026Q4 2026	Enhanced security and reduced fraud risk